SAYAK DUTTA

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Mobile Number: **(+49) 1706627596 (WhatsApp enabled)** | Address: **Germany | Open to relocation and remote roles**

Senior **Full-Stack AI Engineer with 10+ years of experience** designing and scaling production AI systems across LLM platforms, real-time ML pipelines, and distributed infrastructure. Specialized in **RAG architectures, multi-agent systems, and real-time AI applications**, delivering end-to-end platforms from data pipelines and model development to APIs, product interfaces, and production reliability.
Proven track record building **low-latency AI systems used in live operational workflows**, improving system latency by **60%**, reducing infrastructure costs by **45%**, and maintaining **99.9% production reliability**. Experienced collaborating with product and engineering teams to turn AI capabilities into scalable software products.

PROFESSIONAL EXPERIENCE

AI Engineer, **Biss Net GmbH**, Germany ***Nov 2025 – Present***

- Architected and led development of a **multi-agent LLM platform** powering operational workflows for **100+ users**.
- Re-architected retrieval and agent orchestration reducing response latency **12s to <6s**, increasing weekly active usage **38%**.
- Designed **hybrid RAG architecture (HNSW retrieval + semantic re-ranking)** improving context relevance **42%**.
- Implemented hallucination auditing, permission-scoped retrieval, and PII-safe data isolation achieving **99% production reliability**.
- Optimized chunking, retrieval caching, and model routing pipelines reducing **LLM infrastructure costs 45%**.

Senior Machine Learning Engineer, **Torto Labs Pvt Ltd.**, India (Remote) ***Aug 2021 – Sept 2025***

- Led development of **enterprise AI systems powering customer intelligence platforms processing 1M+ records/month**.
- Built distributed **RAG assistant combining vector retrieval, structured data access, and LLM reasoning** for analytics workflows.
- Reduced lookup latency from **60s to <20s**, increasing product adoption **22% to 61%**.
- Designed **model evaluation framework** measuring retrieval recall, grounding accuracy, and latency budgets for production AI releases.
- Architected **event-driven inference pipelines and serverless infrastructure**, maintaining **99.9% system availability**.
- Built CI/CD pipelines and automated deployment workflows for ML systems, significantly reducing deployment time for pipeline updates.

Machine Learning Engineer, **Berylls GmbH**, Germany ***Jul 2023 – Dec 2023***

- Productionized **recommendation models into low-latency APIs serving 100K+ prediction requests/month**.
- Improved recommendation-driven engagement **25%** through feature engineering and model optimization.
- Built **data validation, drift monitoring, and automated retraining pipelines** to maintain model performance in production.
- Containerized ML workflows using **Docker** reducing **MLOps** operational overhead **80%**.

Senior Data Analyst, **Nailbiter Inc.**, India ***Oct 2020 - Feb 2021***

- Built **NLP pipelines and semantic analysis systems** for consumer insights platforms.
- Improved targeting accuracy **20%** and reduced data processing time **30%** through automated pipelines.

Analyst, **eClerx Services Ltd.,** India ***Apr 2018 – Oct 2020***

- Designed **entity resolution and fuzzy matching systems processing 500K+ records** improving matching accuracy **35%**.
- Automated reporting and compliance pipelines reducing manual analysis effort **40%**.

SKILLS

AI / Machine Learning: LLMs, RAG, NLP, Deep Learning, Recommendation Systems, Time-Series Modeling, Feature Engineering, Model Evaluation
Programming: Python, TypeScript / JavaScript, SQL
AI Frameworks: PyTorch, LangChain, Hugging Face, OpenAI API
Full-Stack Development: Node.js, FastAPI, Next.js, REST APIs, WebSockets
Distributed Systems: Kafka, Redis, Vector Databases (FAISS), Event-Driven Pipelines
Infrastructure & MLOps: Docker, Kubernetes, CI/CD, Serverless Architectures, Model Monitoring, Observability

PROJECTS

ResearchRAG - AI Research Paper Digestor & Explorer

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- Engineered a **full-stack RAG system** (FastAPI + FAISS) that processes 30-page academic PDFs into summaries, strengths/weaknesses, and future research directions in under 30 seconds.
- Built an **LLM-powered interactive chat** grounded in embeddings, delivering 95%+ context-relevant answers using free OpenRouter GPT-OSS models.
- Delivered a **production-ready solution** with robust error handling, logging, and seamless frontend–backend integration.
- Designed a **responsive UI** with export options (PDF/Markdown), making research digestion 3–5× faster for researchers and students.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, FAISS, PyMuPDF, OpenRouter GPT-OSS.

AskPostgres - Natural Language to SQL for PostgreSQL

- Built a full-stack application that converts natural language into SQL queries, enabling **business users to query PostgreSQL databases without technical SQL knowledge**.
- Implemented authentication, role-based access control, error handling, and interactive UI with light/dark themes; validated with **extensive query variations** to ensure robustness.
- Improved analyst efficiency by **reducing SQL-writing time by up to 80%**, supporting business analytics, data exploration, and performance monitoring.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, FAISS, PyMuPDF, OpenRouter GPT-OSS.

OutreachPilot - Campaign Automation & Tracking Platform

- Engineered a full-stack automation platform for end-to-end campaign orchestration using **FastAPI**, **Next.js**, **Tailwind**, and **PostgreSQL**, ensuring smooth and reliable workflow execution.
- Integrated automated email delivery via the **Resend API**, with **custom pixel + redirect tracking** to accurately measure open rates, CTR, and variant performance.
- Built modular backend **microservices** with **async workers**, **queues**, and **real-time logging**, delivering high scalability, stability, and fault-tolerant performance.
- Delivered production-ready infrastructure with **Dockerized deployment** and **CI/CD pipelines**, enabling enterprise-grade reliability and rapid feature iteration.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, PostgreSQL, Docker, Resend API

TalkToTube - YouTube Transcript Explorer with LLM Q&Arer

- Built a **full-stack application** (FastAPI, Next.js, Tailwind, PostgreSQL, Docker) that ingests and processes YouTube transcripts into searchable, structured text.
- Implemented an **LLM-powered Q&A system** enabling users to ask natural language questions and receive context-grounded answers from video content.
- Designed a **responsive web interface** with role-based access, real-time results, and export options for summaries and highlights.
- Integrated **vector search (FAISS)** and embeddings for efficient retrieval, delivering fast and accurate results across long-form video content.
- Delivered production-ready infra with **containerized deployment (Docker)** and CI/CD pipelines, ensuring scalability and reproducibility.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, PostgreSQL, FAISS, Docker

EDUCATION

University of Mumbai, Mumbai, India — ***August 2012 – May 2016***

Bachelor of Engineering– Mechanical Engineering | Grade: First Division

Languages: English (Fluent), Hindi (Native), German (A2, actively improving)

Visa: Valid in Germany; open to relocation **Work Authorization:** EU Blue Card eligible | Open to employer-sponsored visas